Quang Van Tran

San Antonio, Texas, USA quangvan.tran@utsa.edu

LinkedIn Google Scholar

Website ORCID

EDUCATION

Ph.D. in Civil Engineering **2025-present**

The University of Texas at San Antonio, The United States

GPA until now: 3.95/4.0

MEng in Architectural Engineering **2021-2023**

Kyung Hee University, Republic of Korea

GPA: 4.12/4.3 (97.70/100)

B.E in Environmental Engineering **2013 – 2017**

Ho Chi Minh City University of Natural Resources and Environment, Vietnam

RESEARCH INTERESTS

The current research interests are:

Urban Climate: Development of numerical simulation, deep learning, and remote-sensing satellite methods for analyzing and predicting urban climates. Research focuses on thermal inequities associated with heat vulnerability, surface urban heat island intensity, social vulnerability, and historical redlining.

Indoor environmental modeling. Developing numerical simulation and machine learning methods to predict indoor environments, including airflow patterns, indoor temperature, and daylighting in buildings.

Optimizing energy consumption. Applying deep reinforcement learning to optimize building energy use while maintaining thermal comfort.

HONORS AND AWARDS

Scholarship Award (Amount: $500), 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings (IAQVEC 2026) at the University of Southern California (USC) in Los Angeles.

Academic Travel Fund (Amount: $750), Graduate School, University of Texas at San Antonio; 2026.

Research Travel Award (Amount: $500), School of Architecture and Planning, University of Texas at San Antonio; 2026.

Fully Funded Ph.D. Scholarship, University of Texas at San Antonio, United States 2025 – 2029

Presidential Scholarship, Kyung Hee University, Republic of Korea 2021 – 2023

Japan – Asia Youth Exchange Program (SAKURA Science Program), Kyushu University 2017

Annual Scholarship for Outstanding Academic Performance, Faculty of Environment, University of Natural Resources and Environment, Vietnam

PUBLICATIONS

Selected Journal Articles

1. **Quang Van Tran,** Dat Tien Doan, Abdulaziz I. Almulhim, Tan Yigitcanlar (2026) "Why Smart Cities Fail Equity and How AI-BIM Integration Can Fix It: From Review to Framework" **(Submitted).**

2. **Quang Van Tran*,** Farzad Hashemi, Fazel Khayatian, Kiran Bhaganagar, Tobi Eniolu Morakinyo (2026). "UrbanDRL-CFD: A Prospective Framework for Integrating Computational Fluid Dynamics and Deep Reinforcement Learning" **(Submitted).**

3. **Quang Van Tran*,** Parisa Najafian, Farzad Hashemi, Tobi Eniolu Morakinyo, Esteban López Ochoa (2025). "Year-Round Heat Exposure and Structural Spatial Vulnerability: A Case Study of San Antonio, Texas", *Sustainable Cities and Society.* https://doi.org/10.1016/j.scs.2025.107021.

4. **Tran Van Quang,** Dat Tien Doan (2024). "Online Transfer Learning for Accelerating Deep Reinforcement Learning for Building Energy Management", *Journal of Building Performance Simulation*. https://doi.org/10.1080/19401493.2025.2511826.

5. **Tran Van Quang,** Dat Tien Doan, Jack Ngarambe, Ali GhaffarianHoseini; Amirhosein Ghaffarianhoseini; Tongrui Zhang (2024) "AI Management Platform for Privacy-Preserving Indoor Air Quality Control: Review and Future Direction," *Journal of Building Engineering.* https://doi.org/10.1016/j.jobe.2024.111712.

6. **Tran Van Quang,** Dat Tien Doan, Nguyen Lu Phuong, Tongrui Zhang, Ali Ghafarianhoseini, Amirhosein Ghaffarianhoseini (2023). "Predicting Indoor Temperature Distribution in Naturally Ventilated Office Room using Long Short-Term Memory Networks," *Architectural Engineering and Design Management.* https://doi.org/10.1080/17452007.2024.2449244.

7. **Tran Van Quang***, Dat Tien Doan, Geun Young Yun (2024). "Recent Advances and

Effectiveness of Machine Learning Models for Fluid Dynamics in the Built Environment," *International Journal of Modelling and Simulation* https://doi.org/10.1080/02286203.2024.2371682.

8. **Tran Van Quang**[*]**,** Dat Tien Doan, Nguyen Lu Phuong, Geun Young Yun (2024). "Data-Driven Prediction of Indoor Airflow Distribution in Naturally Ventilated Residential Buildings using Combined CFD Simulation and Machine Learning (ML) approach," *Journal of Building Physics.* https://doi.org/10.1177/17442591231219025.

9. **Tran Van Quang**[*], Nguyen Lu Phuong (2024). "Using Deep Learning to Optimize the HVAC System in the Residential Building," *Journal of Green Building*. https://doi.org/10.3992/jgb.19.1.29.

[*]*Corresponding author*

Selected Peer-Reviewed Conference Papers

1. **Quang Van Tran**[*a]**,** Farzad Hashemi, Phuong D. Dao (2026) "Surface-Heat Disparities across Historically Redlined U.S. Cities Vary by Climate, Income, and Time of Day", (Submitted), AGU Annual Meeting 2026, San Francisco, CA

2. **Quang Van Tran**[*a], Farzad Hashemi, Parisa, Najafian, Esteban López Ochoa (2026). "Roof Retrofit Scenarios for Thermally Vulnerable Homes: A CFD-Based Evaluation of Indoor Comfort and Air Quality in San Antonio, Texas," *The 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings*, Los Angeles, California, USA, https://doi.org/10.1051/e3sconf/202671610005

3. Farzad Hashemi, Gerald Mills, **Quang Van Tran**[a], Parisa Najafian (2025). "Data-Driven Urban Heat Mitigation: Integrating CFD and Machine Learning for Adaptive Cooling Strategies," *12th International Conference on Urban Climate,* Rotterdam, Netherlands, https://doi.org/10.5194/icuc12-985

4. **Tran Van Quang**, Nguyen Lu Phuong, Dat Tien Doan (2023). "Efficient Prediction of Indoor Airflow in Naturally Ventilated Residential Buildings using a CFD-DNN model approach," *The 2nd International Conference on Green Building, Civil Engineering and Smart City*, China, https://doi.org/10.1007/978-981-99-9947-7_76 (**Indexed in Scopus**).
[a] *Presenter*
[*]*Corresponding author*

PROFESSIONAL SERVICE

GUEST EDITOR

Guest Editor, *Indoor Air* (Wiley) | 2026–2027

Special Issue: Beyond Outdoor Temperature: Housing Conditions, Indoor Heat Exposure, and Human Health **(Special Issue)**

PEER-REVIEWER

Journal:

- Building and Environment
- Environment, Development, and Sustainability
- Numerical Heat Transfer, Part B: Fundamentals
- Numerical Heat Transfer, Part A: Applications
- Alexandria Engineering Journal
- Scientific Report Journal
- Journal of Building Performance Simulation
- Building Simulation
- Discover Computing
- Expert Systems with Applications
- Journal of Building Engineering
- Frontiers of Architectural Research
- Urbanization, Sustainability and Society
- Journal of Electronic Science and Technology
- Discovery Applied Sciences
- Energy Efficiency

International Conferences:

- International Conference of Green+Digital+Intelligent Built Environment (GDI 2025), hosted by Auckland University of Technology, Auckland, New Zealand.

- International Conference of Green+Digital+Intelligent Built Environment (GDI 2026), hosted by Auckland University of Technology, Auckland, New Zealand.

ACADEMIC SERVICE & LEADERSHIP

Program Coordinator & Session Moderator | Research Colloquium **Fall 2026**

School of Architecture and Planning, The University of Texas at San Antonio

Formulated, planned, and executed the school-wide research colloquium series, managing all operational logistics from inception to completion. Strategically invited, vetted, and scheduled graduate researchers to present cutting-edge findings to senior faculty and the broader student body. Served as the lead master of ceremonies and moderator for live research presentations, facilitating technical Q&A sessions and driving cross-disciplinary academic discourse.

MENTORSHIP

- Fatemeh Khajeh Dehaghani, 2025 – Now, M.S. Student at The University of Texas at San Antonio.

PROFESSIONAL MEMBERSHIP

Member of the Sakura Science Club

Member of the American Association of Geographers (AAG)

Member of the American Geophysical Union (AGU)

TECHNICAL SKILLS

Programming languages and mathematical packages: Python.
Computer-aided design/engineering: OpenFOAM, ANSYS Fluent packages, and Rhinoceros Grasshopper packages.

Others: ArcGIS Pro, Google Earth Engine, SPSS, Linux (Ubuntu), OpenLCA.

LANGUAGES

English: Can read, write, speak, understand, and peer review
Vietnamese: Vietnamese (native)